Step 1 · User uploads dataset and invokes the Data plugin

Conversation starts in Claude Cowork

STEP 1 / 10

Cowork · /data:explore-data

You 10:42 AM

/data:explore-data
Read the PDF, understand the tasks for assignment 2a. Use the data plugin. Meet all requirements for assignment 2a. Create nice sharp charts and executive dashboards as needed.

Attached files

- saas_monthly_metrics.csv (480 rows, 12 cols)
- Week 2 Assignment.pdf (Part A — AI Data Analyst)

Claude 10:42 AM

Loading the data:explore-data skill … reading the PDF brief … starting profiling now. I'll do: (1) statistical summary with outliers, (2) single-page dashboard, (3) ten visualizations, (4) ten-slide executive deck.

Step 2 · Data plugin loaded — explore-data skill active

Skill instructions streamed into the working context

STEP 2 / 10

Skill loader · data:explore-data

✓ Skill loaded: data:explore-data

Capabilities: profile, statistical summary, outlier detection, dimension/metric classification

bash · python3

```
# /explore-data — Profile and Explore a Dataset

Workflow:
  1. Access the data (CSV / table / file)
  2. Understand structure  — grain, types, primary key
  3. Generate data profile — null %, distinct counts, summary stats
  4. Identify data quality issues
  5. Discover relationships and patterns
  6. Suggest dimensions, metrics, follow-ups

Statistical summary (numeric columns):
  • count, mean, median (p50), std
  • percentiles: p1, p5, p25, p75, p95, p99
  • outliers via 1.5 × IQR rule

Quality flags:
  · null rate > 5% (warn) / > 20% (alert)
  · suspicious placeholders (N/A, TBD, 999999)
  · low cardinality surprises (e.g. user_id with few values)
  · referential integrity & business-rule violations

>> Loading dataset …
```

Step 3 · Dataset profile and column classification

480 rows × 12 columns · 30 months · 4 regions · 4 plans · 0 nulls · 0 duplicates

STEP 3 / 10

bash · python3

```
$ python3 analysis.py  --profile

=== OVERVIEW ===
  rows               : 480
  columns            : 12
  months covered     : 30
  date range         : Jan-2023 → Jun-2025
  regions            : APAC, Europe, MENA, North America
  plans              : Basic, Pro, Business, Enterprise
  missing values     : 0
  duplicate keys     : 0   (unique on month × region × plan)

=== COLUMN CLASSIFICATION ===
  dimension : month, region, plan
  metric    : new_subscribers, churned_subscribers, active_subscribers
              arpu_usd, mrr_usd, marketing_spend_usd, support_tickets
  ratio     : net_revenue_retention  (range 0.98 – 1.16)
  score     : nps_score              (range 9.2 – 55.2)

=== DATA TYPES ===
  month                 object   (parsed → datetime)
  region / plan         object   (categorical)
  *_subscribers         int64
  *_usd / nrr / nps     float64

Profile complete — proceeding to statistical summary …
```

Step 4 · Statistical summary — mean, median, std, quartiles

Computed for every numeric metric

STEP 4 / 10

statistical_summary.csv

Metric	Mean	Median	Std	Min	Q1 (25%)	Q3 (75%)	Max
new_subscribers	480.0	186.8	222.7	47.2	155.0	219.0	310.0
churned_subscribers	480.0	44.0	90.7	53.4	12.9	89.0	180.0
active_subscribers	480.0	1.9K	2.8K	1.3K	855.1	2.8K	5.0K
arpu_usd	480.0	50.8	162.0	154.2	28.6	96.8	434.4
mrr_usd	480.0	100.6K	468.1K	581.9K	57.2K	224.3K	1.88M
net_revenue_retention	480.0	1.04	1.07	0.04	1.00	1.07	1.14
marketing_spend_usd	480.0	40.7K	45.0K	5.8K	35.8K	45.6K	53.7K
support_tickets	480.0	13.0	24.7	15.4	0.00	24.0	53.0
nps_score	480.0	24.1	30.1	8.73	15.6	29.9	44.7

Step 5 · Outlier detection (1.5 × IQR rule)

Counts and fences for every numeric column

STEP 5 / 10

outliers_summary.csv

Metric	Q1	Q3	IQR	Lower fence	Upper fence	# outliers	% rows
new_subscribers	186.8	253.0	66.2	87.4	352.4	0	0.0%
churned_subscribers	44.0	134.0	90.0	-91.0	269.0	0	0.0%
active_subscribers	1.9K	3.7K	1.8K	-699.8	6.3K	1	0.21%
arpu_usd	50.8	208.1	157.3	-185.2	444.1	6	1.25%
mrr_usd	100.6K	552.3K	451.7K	-576.9K	1.23M	57	11.88%
net_revenue_retention	1.04	1.10	0.06	0.94	1.20	0	0.0%
marketing_spend_usd	40.7K	49.4K	8.7K	27.6K	62.5K	0	0.0%
support_tickets	13.0	35.0	22.0	-20.0	68.0	1	0.21%
nps_score	24.1	35.8	11.8	6.39	53.5	1	0.21%

Interpretation

- MRR shows 57 high outliers and ARPU shows 6 — both driven by Enterprise-tier records.
- These are expected business outliers (premium plan revenue), not data-quality issues.
- NRR, marketing spend, and NPS each have ≤ 1 outlier — distributions are well-bounded.

Step 6 · Headline KPIs computed

Latest-month rollups + growth deltas saved to kpis.json

STEP 6 / 10

TOTAL MRR (JUN-2025) **$14.43M** +4.38% MoM	MRR GROWTH SINCE LAUNCH **+1,144%** 30 months	ACTIVE SUBSCRIBERS **75.6K** 5.2× the Jan-2023 base
MONTHLY CHURN **2.77%** logo churn	NET REVENUE RETENTION **107.9%** expansion above 100%	AVERAGE NPS **41.6** trending up

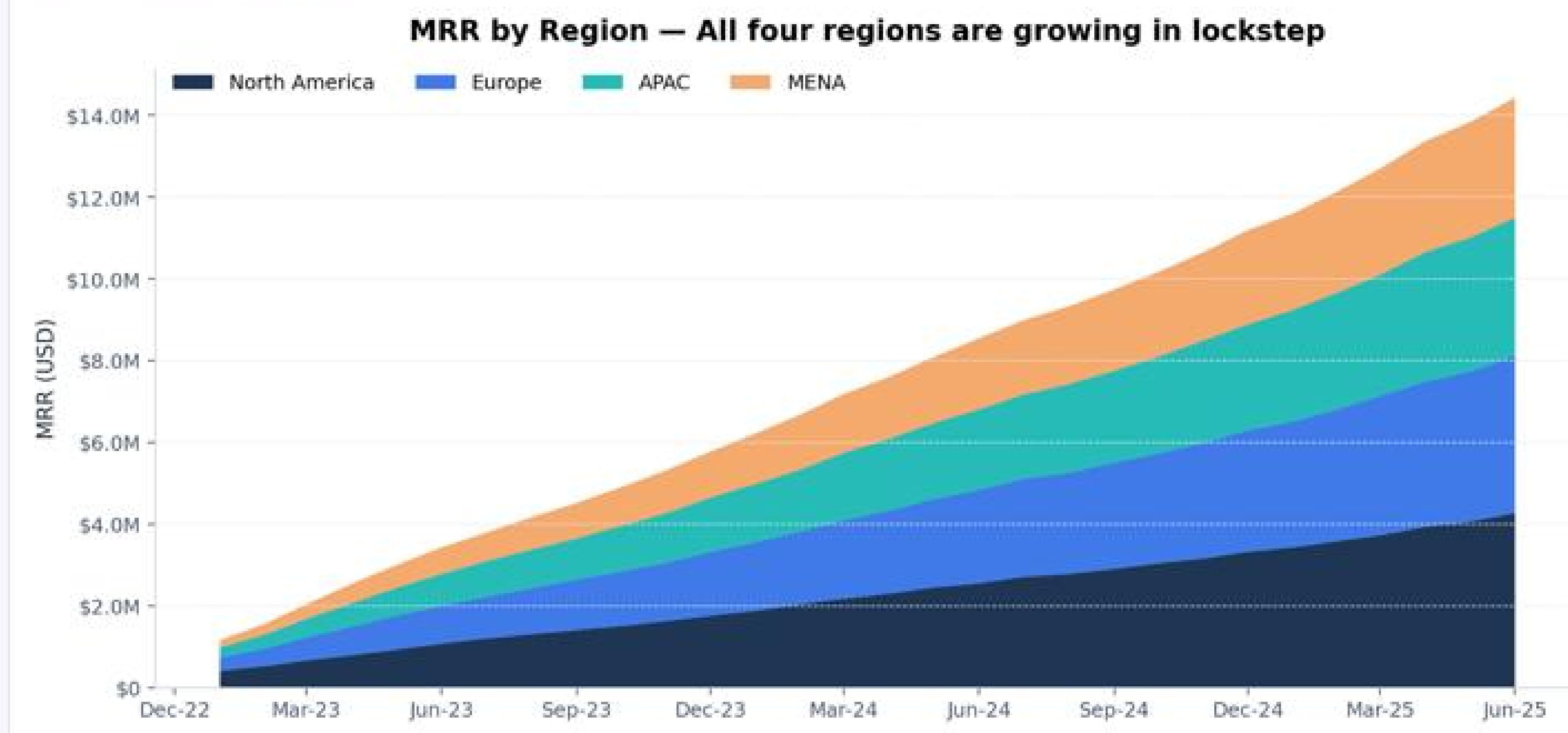
Step 7 · Generating 10 publication-quality visualizations

matplotlib + seaborn · 180 dpi · brand palette

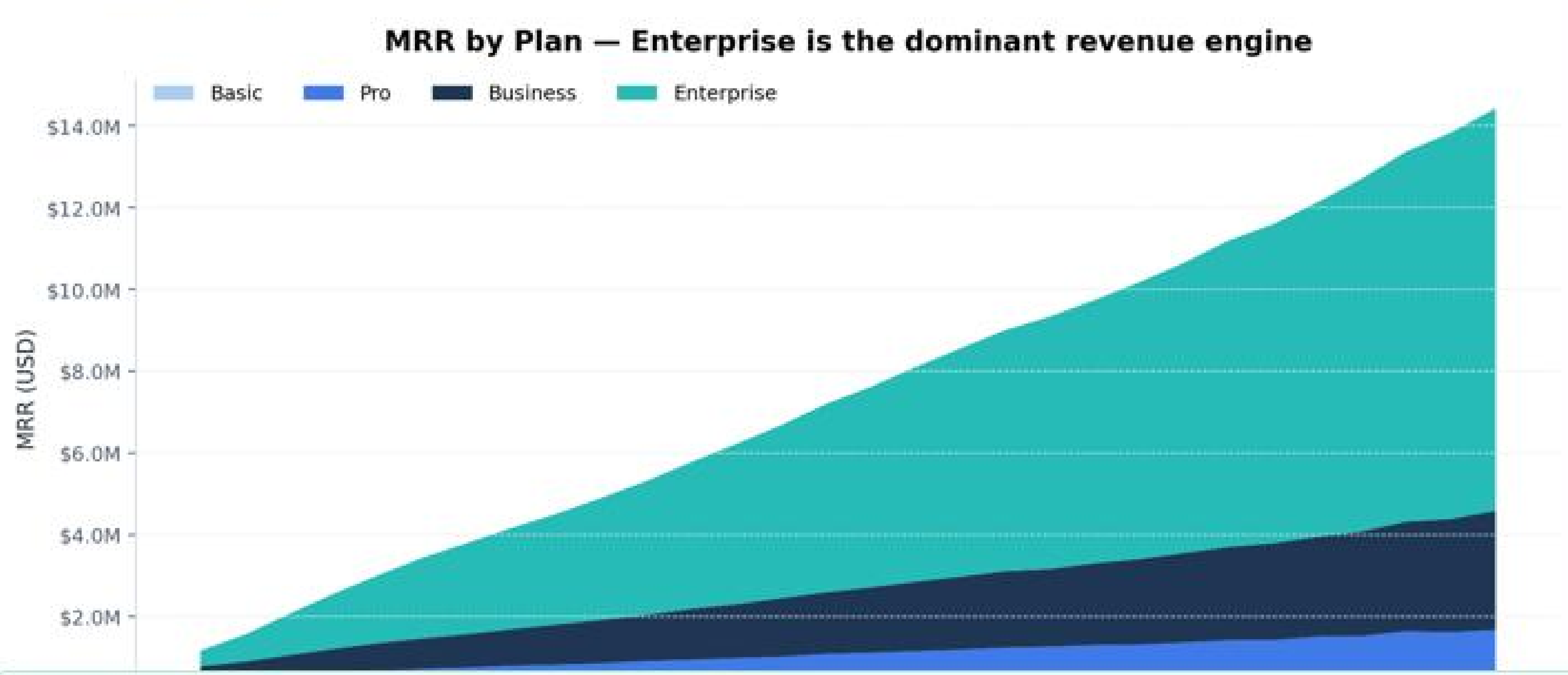
STEP 7 / 10

01 · Total MRR trend

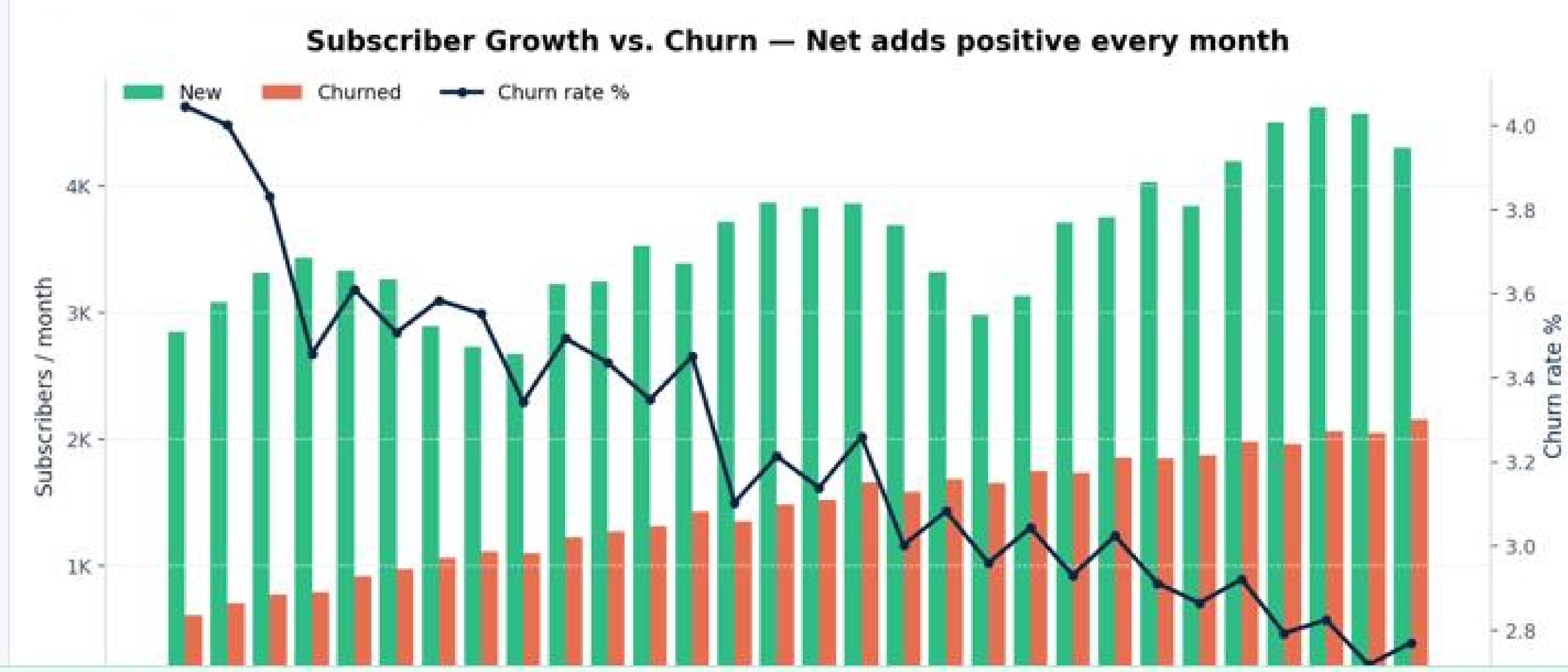
02 · MRR by Region



03 · MRR by Plan



05 · New vs Churned



✓ 10 / 10 charts saved to outputs/charts/

Files: 01_mrr_trend.png · 02_mrr_by_region.png · 03_mrr_by_plan.png · 04_active_subs_by_region.png · 05_new_vs_churn.png · 06_arpu_heatmap.png · 07_nrr_box_by_plan.png · 08_nps_heatmap.png · 09_marketing_efficiency.png · 10_correlation_heatmap.png

Claude · Cowork mode

Data plugin · ✓ active

Step 8 · Single-page executive dashboard assembled

Self-contained HTML — KPI cards + 10 charts + takeaways

STEP 8 / 10

SaaS_Executive_Dashboard.html · Preview

SaaS Executive Dashboard

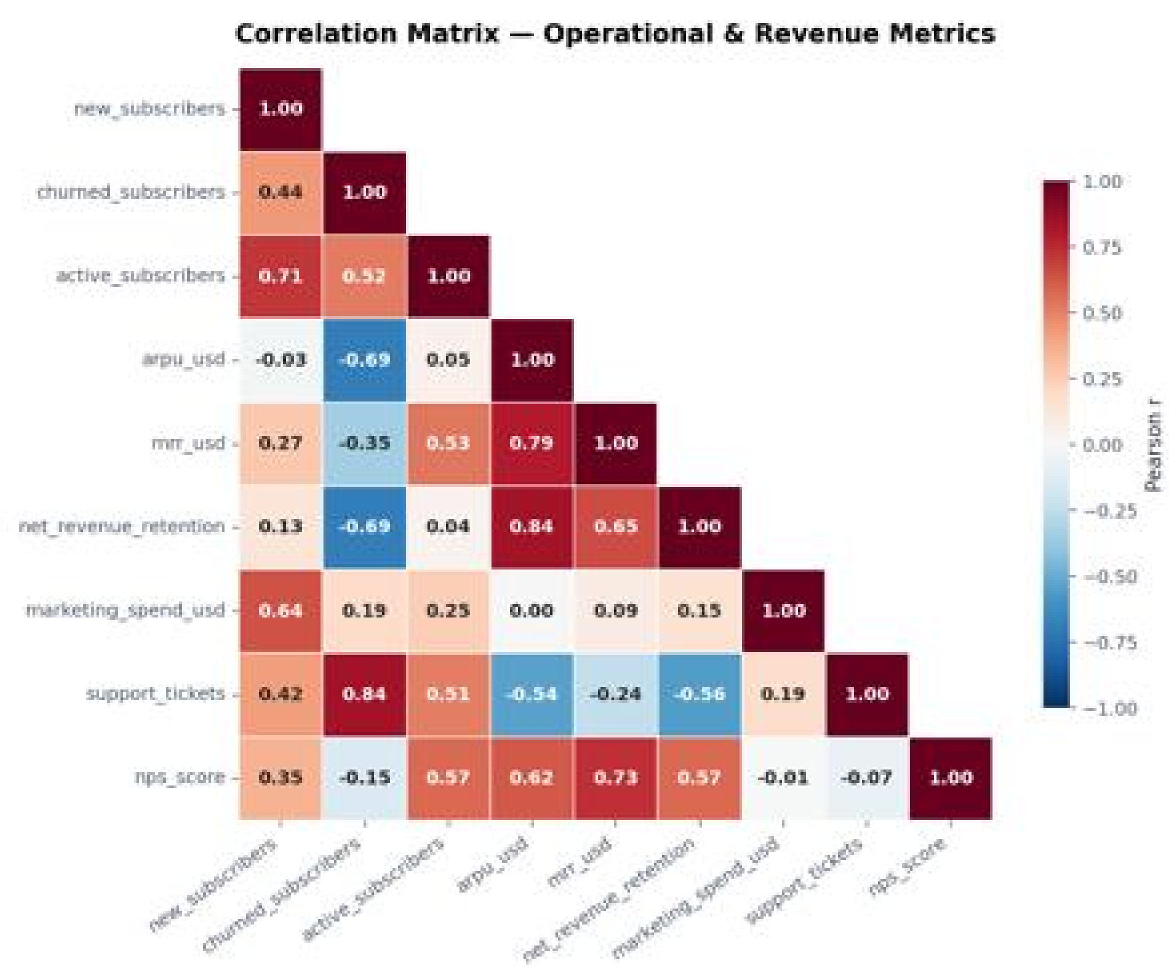
Monthly performance · Jan-2023 → Jun-2025 · 30 months · 4 regions · 4 plan tiers

As of Jun-2025

TOTAL MRR	ACTIVE SUBS	NEW SUBS	CHURN	NRR	NPS
$14.43M	75.6K	4.3K	2.77%	107.9%	41.6

1 · Total MRR Trend

10 · Metric Correlations



KEY TAKEAWAYS FOR LEADERSHIP

Hyper-growth confirmed · Enterprise is the revenue engine · Retention healthy · Support load rising with scale · Marketing efficiency uneven

Claude · Cowork mode

Data plugin · ✓ active

Step 9 · 10-slide executive PowerPoint generated

python-pptx · 16:9 · Midnight Executive palette · QA-passed

STEP 9 / 10

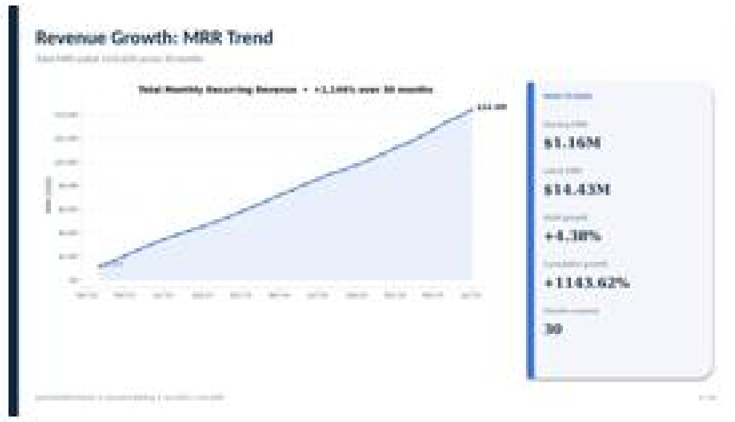
Slide 01

Slide 02

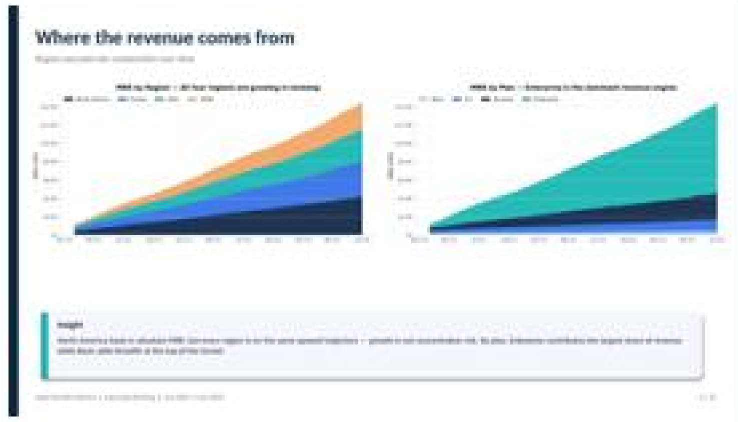
Slide 03

Slide 04

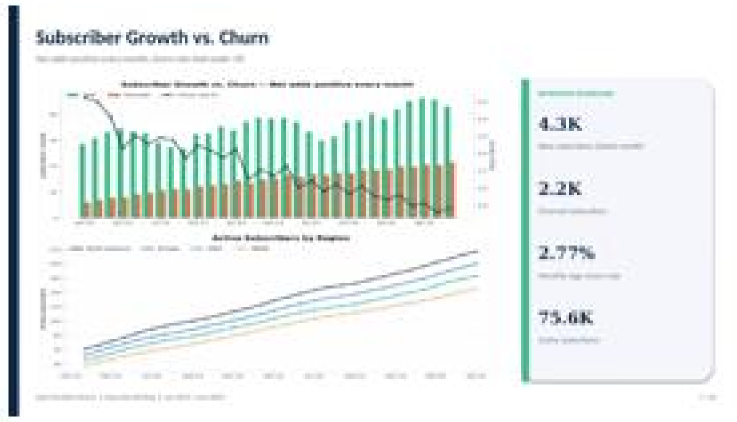
Slide 05



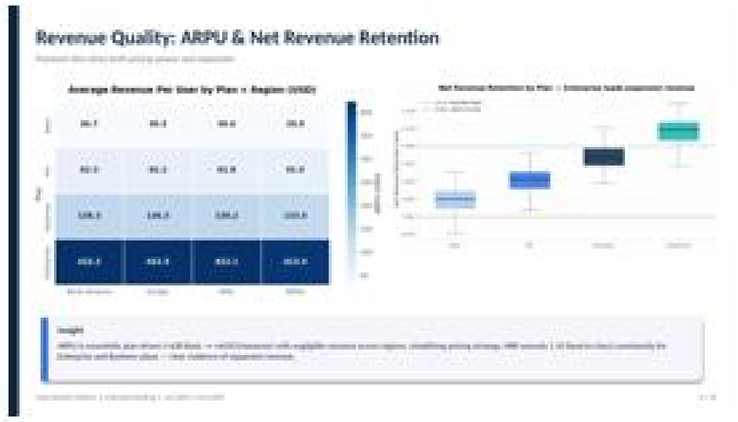
Slide 06



Slide 07

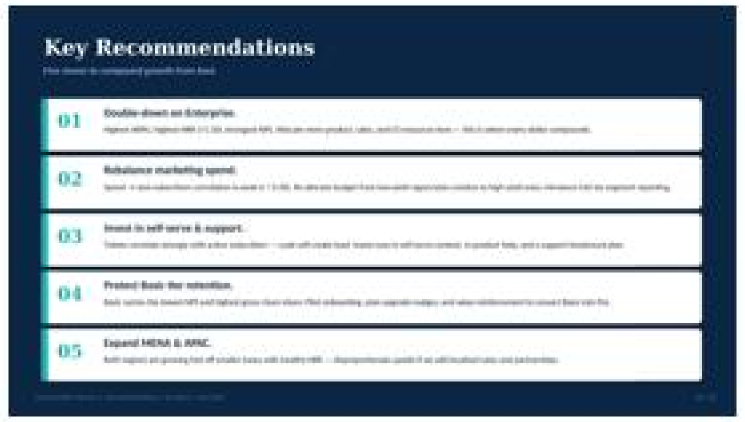


Slide 08



Slide 09

Slide 10



Step 9 of 10 — SaaS_Executive_Briefing.pptx · all 10 slides verified by visual-QA subagent

Step 10 · Deliverables packaged and shared

Files presented in chat with computer:// links

STEP 10 / 10

Cowork · Final reply

Claude 10:58 AM

Here are your Assignment 2A deliverables. Headline numbers: MRR up +1,144% to $14.43M, 75.6K active subs, 2.77% churn, 107.9% NRR, NPS 41.6. Enterprise is the revenue engine; marketing-spend efficiency is the biggest optimization lever.

Files saved to your outputs folder

SaaS_Executive_Briefing.pptx 10-slide executive deck (16:9)	878 KB
SaaS_Executive_Dashboard.html Single-page interactive dashboard	1.5 MB
charts/ (10 PNGs) 01_mrr_trend … 10_correlation_heatmap	1.2 MB
statistical_summary.csv Mean / median / std / quartiles	0.9 KB
outliers_summary.csv IQR fences and outlier counts	0.6 KB
kpis.json Headline KPI rollups (machine-readable)	0.5 KB